

ID No: _____ Name: _____ Sec No.: _____

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

25-02-2026 (Wednesday)

QUIZ-3 (closed book)

B

Max time: 15 Min

Max Marks: 08

Q1. Assume the following initial values: AX = 0005H, BX = 0005H, CX = 0001H. For the snippet of code given below. What is the final value of DX? [1M]

```
CMP AX, BX
JNE L1
MOV DX, 1111H
L1: DEC CX
JCXZ L2
MOV DX, 2222H
JMP L3
L2: MOV DX, 3333H
L3:
.EXIT
END
```

3333H

Q2. For an 8086 processor, Initial values of CX=1000, DX=4040H, AH=20H, AL=F6H, BX=0004H. After the execution of the two instructions given below (Answer the questions in the table). Assume Carry flag(CF) is initially cleared.

```
CBW
MUL BX
```

Question	Solution(in hex)
After execution of the two instructions above, what are the final values of DX and AX (in hexadecimal)?	DX: 0003H AX: FFD8H
Also state whether CF is set or cleared.	SET

Q3. For an x86 processor, answer the questions given in the table for the snippet of the program given in below. Assume all segment registers are initialized to 0859h. SP is pointing to FFAEh

```
.model tiny
.DATA
SAVE_REGS MACRO
    PUSH AX
    PUSH DX
ENDM
.code
.startup
    MOV AX, 2222H
    MOV BX, 2323H
    MOV CX, 3113H
    MOV DX, 4114H
    SAVE_REGS
    PUSH 80H ; Push 8 bit immediate
    PUSH CX
    POP AX
    CALL SUB5
    POP DX
L1: JMP L1
.EXIT
SUB5 PROC NEAR USES BX
    PUSH BP
    MOV BP, SP
    MOV BX, 0AAAAH
    ADD WORD PTR [BP+2], 0002H
    MOV AX, 0BBBBH
    POP BP
    RET
SUB5 ENDP
END
```

Question	Answer(IN HEX)
What is the value of SP just before executing CALL SUB5	FFA8H
What is the value of SP just before executing MOV BP,SP	FFA4
Immediately after MOV BP,SP inside SUB5, what does [BP+2] contain?	Return address
What is the value of DX after the execution of snippet	4114H

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Q1. Assume the following initial values: AX = 0005H, BX = 0005H, CX = 0001H. For the snippet of code given below. What is the final value of DX? [1M]

```
CMP AX, BX
JNE L1
MOV DX, 2222H
L1: DEC CX
JCXZ L2
MOV DX, 3333H
JMP L3
L2: MOV DX, 1111H
L3:
.EXIT
END
```

1111H

Q2. For an 8086 processor, Initial values of CX=1000, DX=4040H, AH=20H, AL=F3H, BX=0004H. After the execution of the two instructions given below (Answer the questions in the table). Assume Overflow flag(OF) is initially cleared.

```
CBW
MUL BX
```

Question	Solution(in hex)
After execution of the two instructions above, what are the final values of DX and AX (in hexadecimal)?	DX: 0003H AX: FFCCH
Also state whether OF is set or cleared.	SET

Q3. For an x86 processor, answer the questions given in the table for the snippet of the program given in below. Assume all segment registers are initialized to 0859h. SP is pointing to BAFEh

```
.model tiny
.DATA
SAVE_REGS MACRO
    PUSH AX
    PUSH DX
ENDM
.code
.startup
    MOV AX, 2222H
    MOV BX, 2323H
    MOV CX, 3113H
    MOV DX, 4114H
    SAVE_REGS
    PUSH 80H ; Push 8 bit immediate
    PUSH CX
    POP AX
    CALL SUB5
    POP DX
L1: JMP L1
.EXIT
SUB5 PROC NEAR USES BX
    PUSH BP
    MOV BP, SP
    MOV BX, 0AAAAH
    ADD WORD PTR [BP+2], 0002H
    MOV AX, 0BBBBH
    POP BP
    RET
SUB5 ENDP
END
```

Question	Answer(IN HEX)
What is the value of SP just before executing CALL SUB5	BAF8H
What is the value of SP just before executing MOV BP,SP	BAF4H
Immediately after MOV BP,SP inside SUB5, what does [BP+2] contain?	Return address
What is the value of DX after the execution of snippet	4114H